ANKIT MATHANKER

Postdoctoral Scholar, University of Chicago / Argonne National Laboratory, Chicago, IL, USA
Email: ankitm@uchicago.edu

Education

- **Ph.D. in Chemical Engineering and Scientific Computing** Aug 2021–Aug 2026
  University of Michigan, USA GPA: 4.0/4.0
- **Master of Science in Chemical Engineering** Jan 2018–Jan 2020
  University of Alberta, Canada GPA: 3.8/4.0
- **Bachelor of Technology in Chemical Engineering** Jul 2013–May 2017
  IIT (ISM) Dhanbad, India GPA: 8.9/10.0

Research Experiences

- **Postdoctoral Scholar**, University of Chicago, USA September 2026–Present
  - Developing covalent organic frameworks based organic cathodes for stable and high capacity lithium ion batteries.
- **Graduate Research Assistant**, University of Michigan, USA Aug 2021–Aug 2026
  - Understanding the effect of electrode potential, electrolyte species, and co-reactants on the electrocatalytic hydrogenation of organics using density functional theory, molecular dynamics.
  - Developing an agentic AI workflow for autonomous transition-state discovery in quantum chemistry simulations, combining multi-agent planning, workflow automation, DFT-based mechanistic modeling. (Ongoing)
- **Graduate Intern**, Lawrence Livermore National Laboratory, USA Jun–Aug 2024
  - Estimating potential-dependent physicochemical properties at metal–electrolyte interfaces using machine learning interatomic potentials developed using MACE architecture.
- **Graduate Research Assistant/Research Assistant**, University of Alberta, Canada Jan 2018–Jan 2021
  - Investigate the synergistic impact of agricultural and forest residue on hydrothermal liquefaction using supercritical water medium in an autoclave reactor.
- **Research Assistant (Indian Academy of Science Fellow)**, IIT Guwahati, India May–Jul 2016
  - Investigate the synthesis of low-cost tubular ceramic membrane (TCM) via the extrusion process. Experimentally and analytically measure water flux and porosity.
- **Undergraduate Dissertation**, IIT (ISM) Dhanbad, India May 2016–Apr 2017
  - Synthesize PEI-impregnated adsorbents for CO_2 capture and simulate adsorbent packed bed for CO_2 adsorption on Aspen Adsorption.

Publications

1. Mathanker, A. Molecular-Level Insights into Organic Adsorption and Reactivity at Electrode-Electrolyte Interfaces. *Thesis* 2026.
2. Madhavan, V.; Mathanker, A.; Sweeney, D.; Oohiro, T.; Goldsmith, B. R. TSAgent: An Agentic Workflow for Autonomous Transition State Search. *https://arxiv.org/abs/2605.14154.*
3. Mathanker, A.; Guo, J.; Goldsmith, B. R.; Varley, J.; Govindarajan, N. Estimating potential-dependent physicochemical properties at metal–electrolyte interfaces using machine learning interatomic potentials. *ACS Electrochemistry*, doi.org/10.1021/acselectrochem.5c00540.
4. de Kam, L. B. T.; Zhu, J. X.; Mathanker, A.; Doblhoff-Dier, K.; Govindarajan, N. Benchmarking short-range machine learning potentials for atomistic simulations of metal/electrolyte interfaces. *https://arxiv.org/abs/2602.22931, (In Print J. Chem. Phys.).*

5. Mathanker, A.; Sharma, G.; Tran, B.; Singh, N.; Goldsmith, B. R. Effect of ions on the aqueous-phase adsorption of small aromatic organics on silver. *J. Phys. Chem. C* 2025, 129, 29, 13433-13444.
6. Mathanker, A.; Halarnkar, S.; Tran, B.; Singh, N.; Goldsmith, B. R. Synergistic effects in organic mixtures for enhanced catalytic hydrogenation and hydrodeoxygenation. *Chem Catalysis* 2024, 4, 101135.
7. Mathanker, A.; Yu, W.; Singh, N.; Goldsmith, B.R. Effects of ions on electrocatalytic hydrogenation and oxidation of organics in aqueous phase. *Curr. Opin. Electrochem.* 40, 101347 (2023).
8. Das, S.; Mathanker, A.; Pudasainee, D.; Khan M.; Kumar, A.; Gupta, R. Synergistic effect of water and co-solvents on the hydrothermal liquefaction of agricultural biomass to produce heavy oil. *International Journal of Energy for a Clean Environment* 2022, 23(4):31-45.
9. Mathanker, A.; Das, S.; Pudasainee, D.; Khan, M.; Gupta, R. A review on hydrothermal liquefaction of biomass for biofuels production with special focus on the effect of process parameters, co-solvents and extraction solvents. *Energies* 2021, 14, 4916.
10. Mathanker, A.; Pudasainee, D.; Kumar, A.; Gupta, R. Hydrothermal liquefaction of lignocellulosic biomass feedstock to produce biofuels: Parameter study and products characterization. *Fuel* 2020, 271, 117534.
11. Mathanker, A. Hydrothermal liquefaction of lignocellulosic biomass to produce biofuels. *Thesis* 2020.

Conference Talks and Posters (most recent)

- Mathanker, A. "Estimating potential-dependent physicochemical properties at metal–electrolyte interface using machine learning interatomic potentials" Gordon Research Conference - AI for Materials, Energy, and Chemical Science, 2026. Texas, USA.
- Mathanker, A.; Guo, J.; Goldsmith, B. R.; Varley, J. B.; Govindarajan, N. "Estimating Water Coverage and Adsorption Isotherms at Electrolyte/Metal Interfaces Using Machine Learning Potentials". AIChE Annual Meeting: Catalysis and Reaction Engineering Division, 2025. Boston, Massachusetts, USA.
- Mathanker, A.; Sharma, G.; Tran, B.; Singh, N.; Goldsmith, B. R. "Effect of ions on the aqueous-phase adsorption of small aromatic organics on silver". AIChE Annual Meeting: Catalysis and Reaction Engineering Division, 2025. Boston, Massachusetts, USA.
- Mathanker, A. "Can aqueous ions modify the adsorption of organics on Ag?" The Student and Postdoc Summer Seminar Series, CSiDIR, 2025, University of Michigan, Ann Arbor, MI, USA.
- Mathanker, A.; Sharma, G.; Tran, B.; Singh, N.; Goldsmith, B. R. "Effect of ions on the aqueous-phase adsorption of small aromatic organics on Ag". The 46th Michigan Catalysis Society Symposium, 2025. Warren, Michigan, USA.

Teaching Experiences

- **Graduate Teaching Assistant**, University of Michigan, USA 2023
  - CHE 538: Statistical thermodynamics
- **Graduate Teaching Assistant**, University of Alberta, Canada 2019
  - CHE 316: Equilibrium stage processes, Spring and Fall terms.
- **Graduate Teaching and Learning (Foundation and Practicum)**, University of Alberta, Canada 2018–2019

Awards

- LLM Hackathon for Applications in Materials and Chemistry–Visionary Awards for RedoxFlow
- CRE Division Travel Award-AIChE AIChE Annual Meeting, USA
- Rackham Travel Award (2022,2023,2024) University of Michigan, USA
- Captain Thomas Farrell Greenhalgh Memorial Graduate Scholarship University of Alberta, Canada

- Mary Louise Imrie Graduate Student Award — University of Alberta, Canada
- Graduate Student Association Travel Award — University of Alberta, Canada
- Summer Research Fellowship Programme Scholarship — Indian Academy of Science, Bengaluru, India

---

Academic/ Non-academic Community services

- Co-chair, Gordon Research Seminar, AI for Materials, Energy and Chemical Sciences, Galveston, Texas, USA, 2028
- Graduate Student Director, AIChE Catalysis and Reaction Engineering Division, 2025-2027
- Graduate Student Representative, Michigan Chapter of North American Catalysis Society, 2022-2024
- Volunteer, UNITEA – Community Social Work Team, University of Alberta, Canada, 2018-2019
- Volunteer, Kartavya – Non-Governmental Organization, IIT (ISM) Dhanbad, India, 2015-2017

Industrial Work Experience

- **Management trainee**, Polyplex Corporation Ltd., India — Jul–Sept 2017
